

AI FitTrack Brainstorming and Ideation

Phase 1

This document explains the problem selection, solution exploration and evidence-based decision to build a secure backend for workout tracking and AI-assisted guidance.

Field	Value
Project	AI FitTrack
Submission	SmartBridge Naan Mudhalvan
Team ID	SWTID-2026-8100
Team size	4
College	DRBCCC Hindu College
Department	B.Sc. Computer Science with Artificial Intelligence
Academic Year	2024–2027
Faculty Guide	Prof. D Saraswathy
Submission Date	
Team Leader	Parthiban M (NM ID: 28538B1DB05060E8D804CDAA88169909, Reg: 222401226)
Team Members	Naveen J (NM ID: 53C2DC1753C6DFDAF8649841B4630C80, Reg: 222401223); Nidesh N S (NM ID: FDE55A8DA9135F6AB7347C01AD1B8A17, Reg: 222401224); Nithish Kumar P (NM ID: BFBD9AD8CEC72229F374877DB82CC0B8, Reg: 222401225)
Project scope	Backend-only REST API
Technology	Node.js, Express.js, MongoDB, Mongoose, JWT and Google Gemini AI
Source baseline	Latest uploaded backend ZIP and supplied verification evidence
Document status	Submission-ready verified technical documentation with complete academic details

Prepared for academic evaluation. Secrets, credentials, private environment files and unverified claims are excluded.

Document contents

1. Problem discovery
2. Empathy and user needs
3. Problem statements
4. Ideas considered and prioritized
5. Selected concept
6. Scope and success criteria
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Problem discovery

Fitness learners may record exercises in notebooks or spreadsheets. That approach makes workout history difficult to organize, search and protect. It also does not transform basic activity data into personalized workout suggestions or progress insights.

AI FitTrack problem to solution map

Backend-only scope derived from the supplied requirements and source code

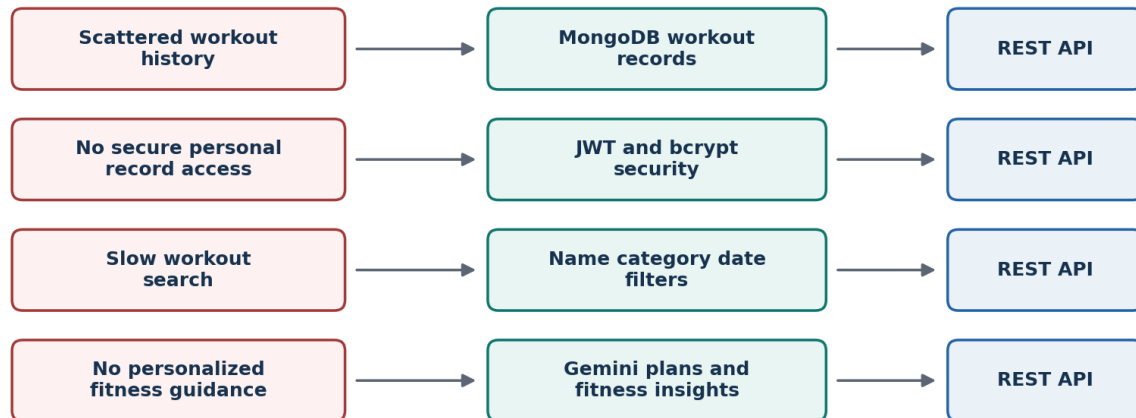


Figure 1 Problem statements mapped to the implemented backend capabilities

Empathy and target user needs

Empathy area	Evidence-based interpretation for an AI FitTrack user
Says	I want one secure place to save and find my workouts.
Thinks	My activity history should help me understand consistency and plan the next workout.
Does	Records workout name, category, duration, calories and date; later searches or updates the record.

Empathy area	Evidence-based interpretation for an AI FitTrack user
Feels	Motivated by clear progress, but concerned about privacy and unreliable generic advice.
Pain	Scattered records, manual calculations, weak search and no personalized guidance.
Gain	Owner-protected history, fast search, structured recommendations and understandable insights.

User need	Observed problem	Backend response
Secure access	Personal records should not be visible to another user.	JWT middleware, fresh user lookup and owner-scoped MongoDB queries.
Reliable history	Manual records are inconsistent and hard to update.	Validated workout documents with CRUD endpoints and timestamps.
Fast discovery	A long history is difficult to browse.	Search by partial name, canonical category and UTC calendar date.
Useful guidance	The user lacks a personalized routine and progress interpretation.	Structured Gemini recommendation and insight endpoints.
Future integration	A frontend or mobile application may be added later.	Standardized REST JSON contracts and CORS configuration.

Problem statements

Persona	Trying to	Obstacle	Need statement
Fitness learner	Maintain a reliable workout history	Manual notes are hard to validate, search and update	A secure API is needed to store and manage consistent workout records.
Fitness learner seeking guidance	Choose a suitable plan and understand progress	Generic advice does not use the user's goal, experience or workout statistics	A structured AI service is needed for general recommendations and evidence-based insights.

Ideas considered

Idea	Strength	Limitation	Decision
Spreadsheet tracker	Quick to start	Weak authentication, search and automation	Rejected

Idea	Strength	Limitation	Decision
Frontend-only local storage app	Simple demonstration	No central database or secure multi-user API	Rejected
Basic workout CRUD API	Clear and practical	No intelligent guidance	Partial option
Secure CRUD plus Gemini backend	Meets tracking, security and personalization goals	Requires external AI configuration	Selected

Idea prioritization

Candidate capability	User value	Feasibility	Priority decision
JWT authentication and owner isolation	High	High	Must implement
Workout CRUD and search	High	High	Must implement
Gemini workout recommendation	High	Moderate	Implement with validation and safe failures
Gemini statistics insights	High	Moderate	Implement for submitted and database statistics
Frontend dashboard	High	Moderate	Future scope because this submission is backend-only
Wearables and nutrition	Moderate	Low for current scope	Future enhancement

Selected concept

AI FitTrack is a backend-only REST API. A user registers, logs in, receives a JWT and then manages only their own workout records. The same authenticated user can request a structured workout recommendation or ask Gemini to interpret submitted statistics or statistics computed from their MongoDB workout history.

Project objectives

- Implement secure registration, login and profile retrieval.
- Protect workout and AI routes using Bearer-token authentication.
- Provide complete workout CRUD and search by name, category and date.
- Store users and workouts in MongoDB through validated Mongoose schemas.
- Generate safe, structured workout recommendations and fitness insights through Google Gemini AI.
- Return predictable JSON success and error responses that can support a future frontend.
- Verify normal, negative, security, retry and ownership behavior.

Scope

Included	Excluded from this backend submission
Authentication, profile, workout CRUD, search, AI recommendation, AI insights, validation, tests and Postman collection	React user interface, charts, nutrition tracking, wearable integration, payment, reminders and production analytics dashboard

Initial success criteria

- Every protected request rejects a missing or invalid JWT.
- A workout is always assigned from the verified token, never from a client-supplied user identifier.
- CRUD and search results remain isolated to the authenticated user.
- Gemini responses follow structured schemas and failures return safe errors.
- No credentials or private environment file are committed to the repository.
- The supplied evidence records 95 automated passes, 17 smoke checks, 19 Atlas checks and 15 live sanitized API records.

Items requiring student confirmation

Pending item	What to provide
Repository	Insert the final public GitHub repository URL after upload.
Deployment	Insert the deployed API base URL if the evaluator requires a hosted demonstration.